

A dynamic illustration of a green and black space submarine, marked with the number '21' on its conning tower, navigating through a blue, turbulent space environment. The submarine is angled towards the viewer, with a bright, glowing white trail of concentric circles behind it, suggesting high-speed travel or a warp effect. In the background, a dark, jagged celestial body looms on the left, and a small satellite or probe is visible in the upper right. The overall scene conveys a sense of action and futuristic exploration.

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WORLDS OF SCIENCE FICTION

FEBRUARY

35 CENTS

A New Space Thriller

AVOIDANCE SITUATION

by James McConnell

POUL ANDERSON • JAMES BLISH • JOHN JAKES • JACK LEWIS
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*** START OF THE PROJECT GUTENBERG EBOOK CATALYSIS ***

CATALYSIS

BY POUL ANDERSON

*Man is a kind of turtle. Wherever
he goes, he will always carry a
shell holding warmth and air—and
with them his human failings....*

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When you looked outside, it was into darkness.

Going out yourself, you could let your eyes accommodate. At high noon, the sun was a sharp spark in a dusky heaven, and its light amounted to about one-ninth of one percent of what Earth gets. The great fields of ice and frozen gases reflected enough to help vision, but upthrust crags and cliffs of naked rock were like blackened teeth.

Seventy hours later, when Triton was on the other side of the primary that it always faced, there was a midnight thick enough to choke you. The stars flashed and glittered, a steely twinkle through a gaunt atmosphere mostly hydrogen—strange, to see the old lost constellations of Earth, here on the edge of the deep. Neptune was at the full, a giant sprawling across eight degrees of sky, bluish gray and smoky banded, but it caught so little sunlight that men groped in blindness. They set up floodlights, or had lamps glaring from their tracs, to work at all.

But nearly everything went on indoors. Tunnels connected the various buildings on the Hill, instruments were of necessity designed to operate in the open without needing human care, men rarely had occasion to go out any more. Which was just as well, for it takes considerable power and insulation to keep a man alive when the temperature hovers around 60 degrees Kelvin.

And so you stood at a meter-thick port of insulglas, and looked out, and saw only night.

Thomas Gilchrist turned away from the view with a shudder. He had always hated cold, and it was as if the bitterness beyond the lab-dome had seeped in to touch him. The cluttered gleam of instruments in the room, desk piled high with papers and microspools, the subdued chatter of a computer chewing a problem, were comforting.

He remembered his purpose and went with a long low-gravity stride to check the mineralogical unit. It was busily breaking down materials fetched in by the robosamplers, stones never found on Earth—because Earth is not the Mercury-sized satellite of an outer planet, nor has it seen some mysterious catastrophe in an unknown time back near the beginning of

things. Recording meters wavered needles across their dials, data tapes clicked out, he would soon have the basic information. Then he would try to figure out how the mineral could have been formed, and give his hypothesis to the computer for mathematical analysis of possibility, and start on some other sample.

For a while Gilchrist stood watching the machine. A cigaret smoldered forgotten between his fingers. He was a short, pudgy young man, with unkempt hair above homely features. Pale-blue eyes blinked nearsightedly behind contact lenses, his myopia was not enough to justify surgery. Tunic and slacks were rumpled beneath the gray smock.

Behold the bold pioneer! he thought. His self-deprecating sarcasm was mildly nonsane, he knew, but he couldn't stop—it was like biting an aching tooth. Only a dentist could fix the tooth in an hour, while a scarred soul took years to heal. It was like his eyes, the trouble wasn't bad enough to require long expensive repair, so he limped through life.

Rafael Alemán came in, small and dark and cheerful. "Allo," he said. "How goes it?" He was one of the Hill's organic chemists, as Gilchrist was the chief physical chemist, but his researches into low-temperature properties were turning out so disappointingly that he had plenty of time to annoy others. Nevertheless, Gilchrist liked him, as he liked most people.

"So-so. It takes time."

"Time we have enough of, *mi amigo*," said Alemán. "Two years we 'ave been here, and three years more it will be before the ship comes to relieve us." He grimaced. "Ah, when I am back to Durango Unit, how fast my savings will disappear!"

"You didn't have to join the Corps, and you didn't have to volunteer for Triton Station," Gilchrist pointed out.

The little man shrugged, spreading slender hands. "Confidential, I will tell you. I had heard such colorful tales of outpost life. But the only result is that I am now a married man—not that I have anything but praise for my dear Mei-Hua, but it is not the abandonment one had hoped for."

Gilchrist chuckled. Outer-planet stations did have a slightly lurid reputation, and no doubt it had been justified several years ago.

After all—The voyage was so long and costly that it could not be made often. You established a self-sufficient colony of scientists and left it there to carry on its researches for years at a time. But self-sufficiency includes psychic elements, recreation, alcohol, entertainment, the opposite sex. A returning party always took several children home.

Scientists tended to be more objective about morals, or at least more tolerant of the other fellow's, than most; so when a hundred or so people were completely isolated, and ordinary amusements had palled, it followed that there would be a good deal of what some would call sin.

"Not Triton," said Gilchrist. "You forget that there's been another cultural shift in the past generation—more emphasis on the stable family. And I imagine the Old Man picked his gang with an eye to such attitudes. Result—the would-be rounders find themselves so small a minority that it has a dampening effect."

"Sí. I know. But you 'ave never told me your real reason for coming here, Thomas."

Gilchrist felt his face grow warm. "Research," he answered shortly. "There are a lot of interesting problems connected with Neptune."

Alemán cocked a mildly skeptical eyebrow but said nothing. Gilchrist wondered how much he guessed.

That was the trouble with being shy. In your youth, you acquired bookish tastes; only a similarly oriented wife would do for you, so you didn't meet many women and didn't know how to behave with them anyhow. Gilchrist, who was honest with himself, admitted he'd had wistful thoughts about encountering the right girl here, under informal conditions where—

He had. And he was still helpless.

Suddenly he grinned. "I'll tell you what," he said. "I also came because I don't like cold weather."

"Came to *Neptune*?"

"Sure. On Earth, you can stand even a winter day, so you have to. Here, since the local climate would kill you in a second or two, you're always well protected from it." Gilchrist waved at the viewport. "Only I wish they didn't

have that bloody window in my lab. Every time I look out, it reminds me that just beyond the wall nitrogen is a solid."

"*Yo comprendo*," said Alemán. "The power of suggestion. Even now, at your words, I feel a chill."

Gilchrist started with surprise. "You know, somehow I have the same—Just a minute." He went over to a workbench. His inframicrothermometer had an air thermometer attached to make temperature corrections.

"What the devil," he muttered. "It is cooled off. Only 18 degrees in here. It's supposed to be 21."

"Some fluctuation, in temperature as in ozone content and humidity," reminded Alemán. "That is required for optimum health."

"Not this time of day, it shouldn't be varying." Gilchrist was reminded of his cigaret as it nearly burned his fingers. He stubbed it out and took another and inhaled to light it.

"I'm going to raise Jahangir and complain," he said. "This could play merry hell with exact measurements."

Alemán trotted after him as he went to the door. It was manually operated, and the intercoms were at particular points instead of every room. You had to forego a number of Earthside comforts here.

There was a murmuring around him as he hurried down the corridor. Some doors stood open, showing the various chemical and biological sections. The physicists had their own dome, on the other side of the Hill, and even so were apt to curse the stray fields generated here. If they had come this far to get away from solar radiations, it was only reasonable, as anyone but a chemist could see, that—

The screen stood at the end of the hall, next to the tunnel stairs. Gilchrist checked himself and stood with a swift wild pulse in his throat. Catherine Bardas was using it.

He had often thought that the modern fashion of outbreeding yielded humans more handsome than any pure racial type could be. When a girl was half Greek and half Amerind, and a gifted biosynthesizer on top of it, a man like him could only stare.

Mohammed Jahangir's brown, bearded face registered more annoyance than admiration as he spoke out of the screen. "Yes. Dr. Bardas," he said with strained courtesy. "I know. My office is being swamped with complaints."

"Well, what's the trouble?" asked the girl. Her voice was low and gentle, even at this moment.

"I'm not sure," said the engineer. "The domes' temperature is dropping, that's all. We haven't located the trouble yet, but it can't be serious."

"All I'm asking," said Catherine Bardas patiently, "is how much longer this will go on and how much lower it's going to get. I'm trying to synthesize a cell, and it takes precisely controlled conditions. If the air temperature drops another five degrees, my thermostat won't be able to compensate."

"Oh, well ... I'm sure you can count on repair being complete before that happens."

"All right," said Catherine sweetly. "If not, though, I'll personally bung you out the main air-lock *sans* spacesuit."

Jahangir laughed and cut off. The light of fluorotubes slid blue-black off the girl's shoulder-length hair as she turned around. Her face was smooth and dark, with high cheekbones and a lovely molding of lips and nose and chin.

"Oh—hello, Tom," she smiled. "All through here."

"Th-th-th—Never mind," he fumbled. "I was only g-going to ask about it myself."

"Well—" She yawned and stretched with breathtaking effect. "I suppose I'd better get back and—"

"Ah, why so, señorita?" replied Alemán. "If the work does not need your personal attention just now, come join me in a leetle drink. It is near dinnertime anyhow."

"All right," she said. "How about you, Tom?"

He merely nodded, for fear of stuttering, and accompanied them down the stairs and into the tunnel. Half of him raged at his own timidity—why hadn't he made that suggestion?

The passages connecting the domes were all alike, straight featureless holes lined with plastic. Behind lay insulation and the pipes of the common heating system, then more insulation, finally the Hill itself. That was mostly porous iron, surprisingly pure though it held small amounts of potassium and aluminum oxides. The entire place was a spongy ferrous outcropping. But then, Triton was full of geological freaks.

"How goes your work?" asked Alemán sociably.

"Oh, pretty well," said Catherine. "I suppose you know we've synthesized virus which can live outside. Now we're trying to build bacteria to do the same."

On a professional level, Gilchrist was not a bad conversationalist. His trouble was that not everyone likes to talk shop all the time. "Is there any purpose in that, other than pure research to see if you can do it?" he inquired. "I can't imagine any attempt ever being made to colonize this moon."

"Well, you never know," she answered. "If there's ever any reason for it, oxide-reducing germs will be needed."

"As well as a nuclear heating system for the whole world, and—What do your life forms use for energy, though? Hardly enough sunlight, I should think."

"Oh, but there is, for the right biochemistry with the right catalysts—analogue to our own enzymes. It makes a pretty feeble type of life, of course, but I hope to get bacteria which can live off the local ores and frozen gases by exothermic reactions. Don't forget, when it's really cold a thermal engine can have a very high efficiency; and all living organisms are thermal engines of a sort."

They took the stairs leading up into the main dome: apartments, refectories, social centers, and offices. Another stair led downward to the central heating plant in the body of the Hill. Gilchrist saw an engineer going that way with a metering kit and a worried look.

The bar was crowded, this was cocktail hour for the swing shift and—popular opinion to the contrary—a scientist likes his meals regular and only lives off sandwiches brought to the lab when he must. They found a table

and sat down. Nobody had installed dial units, so junior technicians earned extra money as waiters. One of them took their orders and chits.

The ventilators struggled gallantly with the smoke. It hazed the murals with which some homesick soul had tried to remember the green Earth. A couple of astronomers at the next table were noisily disputing theories.

"—Dammit, Pluto's got to be an escaped satellite of Neptune. Look at their orbits ... and Pluto is where Neptune should be according to Bode's Law."

"I know. I've heard that song before. I suppose you favor the Invader theory?"

"What else will account for the facts? A big planet comes wandering in, yanks Neptune sunward and frees Pluto; but Neptune captures a satellite of the Invader. Triton's got to be a captured body, with this screwy retrograde orbit. And Nereid—"

"Have you ever analyzed the mechanics of that implausible proposition? Look here—" A pencil came out and began scribbling on the long-suffering table top.

Catherine chuckled. "I wonder if we'll ever find out," she murmured.

Gilchrist rubbed chilled fingers together. Blast it, the air was still cooling off! "It'd be interesting to land a ship on Nep himself and check the geology," he said. "A catastrophe like that would leave traces."

"When they can build a ship capable of landing on a major planet without being squeezed flat by the air pressure, that'll be the day. I think we'll have to settle for telescopes and spectroscopes for a long, long time to come—"

The girl's voice trailed off, and her dark fine head poised. The loudspeaker was like thunder.

"DR. VESEY! DR. VESEY! PLEASE CONTACT ENGINEERING OFFICE! DR. VESEY, PLEASE CONTACT DR. JAHANGIR! OVER."

For a moment, there was silence in the bar.

"I wonder what the trouble is," said Alemán.

"Something to do with the heating plant, I suppose—" Again Catherine's tones died, and they stared at each other.

The station was a magnificent machine; it represented an engineering achievement which would have been impossible even fifty years ago. It kept a hundred human creatures warm and moist, it replenished their air and synthesized their food and raised a wall of light against darkness. But it had not the equipment to call across nearly four and a half billion kilometers of vacuum. It had no ship of its own, and the great Corps vessel would not be back for three years.

It was a long way to Earth.

Dinner was a silent affair that period. There were a few low-voiced exchanges, but they only seemed to deepen the waiting stillness.

And the cold grew apace. You could see your breath, and your thin garments were of little help.

The meal was over, and the groups of friends were beginning to drift out of the refectory, when the intercoms woke up again. This chamber had a vision screen. Not an eye stirred from Director Samuel Vesey as he looked out of it.

His lips were firm and his voice steady, but there was a gleam of sweat on the ebony skin—despite the cold. He stared directly before him and spoke:

"Attention, all personnel. Emergency situation. Your attention, please."

After a moment, he seemed to relax formality and spoke as if face to face. "You've all noticed our trouble. Something has gone wrong with the heating plant, and Dr. Jahangir's crew haven't located the trouble so far.

"Now there's no reason for panic. The extrapolated curve of temperature decline indicates that, at worst, it'll level off at about zero Centigrade. That won't be fun, but we can stand it till the difficulty has been found. Everyone is advised to dress as warmly as possible. Food and air plant crews are going on emergency status. All projects requiring energy sources are cancelled till further notice.

"According to the meters, there's nothing wrong with the pile. It's still putting out as much heat as it always has. But somehow, that heat isn't

getting to us as it should. The engineers are checking the pipes now.

"I'll have a stat of the findings made up and issued. Suggestions are welcome, but please take them to my office—the engineers have their own work to do. Above all, don't panic! This is a nuisance, I know, but there's no reason to be afraid.

"All personnel not needed at once, stand by. The following specialists please report to me—"

He read off the list, all physicists, and closed his talk with a forced grin and thumbs up.

As if it had broken a dam, the message released a babble of words. Gilchrist saw Catherine striding out of the room and hastened after her.

"Where are you going?" he asked.

"Where do you think?" she replied. "To put on six layers of clothes."

He nodded. "Best thing. I'll come along, if I may—my room's near yours."

A woman, still in her smock, was trying to comfort a child that shivered and cried. A Malayan geologist stood with teeth clattering in his jaws. An engineer snarled when someone tried to question him and ran on down the corridor.

"What do you think?" asked Gilchrist inanely.

"I don't have any thoughts about the heating plant," said Catherine. Her voice held a thin edge. "I'm too busy worrying about food and air."

Gilchrist's tongue was thick and dry in his mouth. The biochemistry of food creation and oxygen renewal died when it got even chilly.

Finished dressing, they looked at each other in helplessness. Now what?

The temperature approached its minimum in a nosedive. There had always been a delicate equilibrium; it couldn't be otherwise, when the interior of the domes was kept at nearly 240 degrees above the surrounding world. The

nuclear pile devoted most of its output to maintaining that balance, with only a fraction going to the electric generators.

Gilchrist thrust hands which were mottled blue with cold into his pockets. Breath smoked white before him. Already a thin layer of hoarfrost was on ceiling and furniture.

"How long can we stand this?" he asked.

"I don't know," said Catherine. "Not too long, I should think, since nobody has adequate clothes. The children should ... suffer ... pretty quickly. Too much drain on body energy." She clamped her lips together. "Use your mental training. You can ignore this till it begins actually breaking down your physique."

Gilchrist made an effort, but couldn't do it. He could stop shivering, but the chill dank on his skin, and the cold sucked in by his nose, were still there in his consciousness, like a nightmare riding him.

"They'll be dehumidifying the air," said Catherine. "That'll help some." She began walking down the hall. "I want to see what they're doing about the food and oxy sections."

A small mob had had the same idea. It swirled and mumbled in the hall outside the service rooms. A pair of hard-looking young engineers armed with monkey wrenches stood guard.

Catherine wormed her way through the crowd and smiled at them. Their exasperation dissolved, and one of them, a thickset red-head by the name of O'Mallory, actually grinned. Gilchrist, standing moodily behind the girl, could hardly blame him.

"How's it going in there?" she asked.

"Well, now, I suppose the Old Man *is* being sort of slow about his bulletins," said O'Mallory. "It's under control here."

"But what are they doing?"

"Rigging electric heaters, of course. It'll take all the juice we have to maintain these rooms at the right temperature, so I'm afraid they'll be cutting off light and power to the rest of the Hill."

She frowned. "It's the only thing, I suppose. But what about the people?"

"They'll have to jam together in the refectories and clubrooms. That'll help keep 'em warm."

"Any idea what the trouble is?"

O'Mallory scowled. "We'll get it fixed," he said.

"That means you don't know." She spoke it calmly.

"The pile's all right," he said. "We telemetered it. I'd'a done that myself, but you know how it is—" He puffed himself up a trifle. "They need a couple husky chaps to keep the crowd orderly. Anyhow, the pile's still putting out just as it should, still at 500 degrees like it ought to be. In fact, it's even a bit warmer than that; why, I don't know."

Gilchrist cleared his throat. "Th-th-then the trouble is with the ... heating pipes," he faltered.

"How did you ever guess?" asked O'Mallory with elaborate sarcasm.

"Lay off him," said Catherine. "We're all having a tough time."

Gilchrist bit his lip. It wasn't enough to be a tongue-tied idiot, he seemed to need a woman's protection.

"Trouble is, of course," said O'Mallory, "the pipes are buried in insulation, behind good solid plastic. They'll be hard to get at."

"Whoever designed this farce ought to have to live in it," said his companion savagely.

"The same design's worked on Titan with no trouble at all," declared O'Mallory.

Catherine's face took on a grimness. "There never was much point in making these outer-planet domes capable of quick repair," she said. "If something goes wrong, the personnel are likely to be dead before they can fix it."

"Now, now, that's no way to talk," smiled O'Mallory. "Look, I get off duty at 0800. Care to have a drink with me then?"

Catherine smiled back. "If the bar's operating, sure."

Gilchrist wandered numbly after her as she left.

The cold gnawed at him. He rubbed his ears, unsure about frostbite. Odd how fast you got tired—It was hard to think.

"I'd better get back to my lab and put things away before they turn off the electricity to it," he said.

"Good idea. Might as well tidy up in my own place." Something flickered darkly in the girl's eyes. "It'll take our minds off—"

Off gloom, and cold, and the domes turned to blocks of ice, and a final night gaping before all men. Off the chasm of loneliness between the Hill and the Earth.

They were back in the chemical section when Alemán came out of his lab. The little man's olive skin had turned a dirty gray.

"What is it?" Gilchrist stopped, and something knotted hard in his guts.

"*Madre de Díos*—" Alemán licked sandy lips. "We are finished."

"It's not that bad," said Catherine.

"You do not understand!" he shrieked. "Come here!"

They followed him into his laboratory. He mumbled words about having checked a hunch, but it was his hands they watched. Those picked up a Geiger counter and brought it over to a wall and traced the path of a buried heating pipe.

The clicking roared out.

"Beta emission," said Gilchrist. His mouth felt cottony.

"How intense?" whispered Catherine.

Gilchrist set up an integrating counter and let it run for a while. "Low," he said. "But the dosage is cumulative. A week of this, and we'll begin to show the effects. A month, and we're dead."

"There's always some small beta emission from the pipes," said the girl. "A little tritium gets formed down in the pile room. It's ... never been enough to matter."

"Somehow, the pile's beginning to make more H-3, then." Gilchrist sat down on a bench and stared blankly at the floor.

"The laws of nature—" Alemán had calmed down a bit, but his eyes were rimmed with white.

"Yes?" asked Catherine when he stopped. She spoke mostly to fend off the silence.

"I 'ave sometimes thought ... what we know in science is so leetle. It may be the whole universe, it has been in a ... a most improbable state for the past few billion years." Alemán met her gaze as if pleading to be called a liar. "It may be that what we thought to be the laws of nature, those were only a leetle statistical fluctuation."

"And now we're going back onto the probability curve?" muttered Gilchrist. He shook himself. "No, damn it. I won't accept that till I must. There's got to be some rational explanation."

"Leakage in the pipes?" ventured Catherine.

"We'd know that. Nor does it account for the radiation. No, it's—" His voice twisted up on him, and he groped out a cigaret. "It's something natural."

"What is natural?" said Alemán. "How do we know, leetle creeping things as we are, living only by the grace of God? We 'ave come one long way from home." His vision strayed to the viewport with a kind of horror.

Yes, thought Gilchrist in the chilled darkness of his mind, yes, we have come far. Four and a half billion kilometers further out from the sun. The planet-sized moon of a world which could swallow ours whole without noticing. A thin hydrogen atmosphere, glaciers of nitrogen which turn to rivers when it warms up, ammonia snow, and a temperature not far above absolute zero. What do we know? What is this arrogance of ours which insists that the truth on Earth is also the truth on the rim of space?

No!

He stood up, shuddering with cold, and said slowly: "We'd better go see Dr. Vesey. He has to know, and maybe they haven't thought to check the radiation. And then—"

Catherine stood waiting.

"Then we have to think our way out of this mess," he finished lamely. "Let's, uh, start from the beginning. Think back how th-th-the heating plant works."

Down in the bowels of the Hill was a great man-made cave. It had been carved out of the native iron, with rough pillars left to support the roof; walls and ceiling were lined with impermeable metal, but the floor was in its native state—who cared if there was seepage downward?

The pile sat there, heart and life of the station.

It was not a big one, just sufficient to maintain man on Triton. Part of its energy was diverted to the mercury-vapor turbines which furnished electricity. The rest went to heat the domes above.

Now travel across trans-Jovian spaces is long and costly; even the smallest saving means much. Very heavy insulation against the haze of neutrons which the pile emitted could scarcely be hauled from Earth, nor had there been any reason to spend time and labor manufacturing it on Triton.

Instead, pumps sucked in the hydrogen air and compressed it to about 600 atmospheres. There is no better shield against high-energy neutrons; they bounce off the light molecules and slow down to a speed which makes them perfectly harmless laggards which don't travel far before decaying into hydrogen themselves. This, as well as the direct radiation of the pile, turned the room hot—some 500 degrees.

So what was more natural than that the same hydrogen should be circulated through pipes of chrome-vanadium steel, which is relatively impenetrable even at such temperatures, and heat the domes?

There was, of course, considerable loss of energy as the compressed gas seeped through the Hill and back into the satellite's atmosphere. But the pumps maintained the pressure. It was not the most efficient system which could have been devised; it would have been ludicrous on Earth. But on Triton, terminal of nowhere, men had necessarily sacrificed some engineering excellence to the stiff requirements of transportation and labor.

And after all, it had worked without a hitch for many years on Saturn's largest moon. It had worked for two years on Neptune's—

Samuel Vesey drummed on his desk with nervous fingers. His dark countenance was already haggard, the eyes sunken and feverish.

"Yes," he said. "Yes, it was news to me."

Jahangir put down the counter. The office was very quiet for a while.

"Don't spread the word," said Vesey. "We'll confine it to the engineers. Conditions are bad enough without a riot breaking loose. We can take several days of this radiation without harm, but you know how some people are about it."

"You've not been very candid so far," snapped Catherine. "Just exactly what have you learned?"

Jahangir shrugged. There was a white frost rimming his beard. "There've been no bulletins because there's no news," he replied. "We checked the pile. It's still putting out as it should. The neutron flux density is the same as ever. It's the gas there and in our pipes which has gotten cold and ... radioactive."

"Have you looked directly in the pile room—actually entered?" demanded Alemán.

Jahangir lifted his shoulders again. "My dear old chap," he murmured. "At a temperature of 500 and a pressure of 600?" After a moment, he frowned. "I do have some men modifying a trac so it could be driven in there for a short time. But I don't expect to find anything. It's mostly to keep them busy."

"How about the pipes, then?" asked Gilchrist.

"Internal gas pressure and velocity of circulation is just about what it always has been. According to the meters, anyway, which I don't think are lying. I don't want to block off a section and rip it out except as a last resort. It would just be wasted effort, I'm sure." Jahangir shook his turbanned head.

"No, this is some phenomenon which we'll have to think our way through, not bull through."

Vesey nodded curtly. "I suggest you three go back to the common rooms," he said. "We'll be shunting all the power to food and oxy soon. If you have any further suggestions, pass them on ... otherwise, sit tight."

It was dismissal.

The rooms stank.

Some ninety human beings were jammed together in three long chambers and an adjacent kitchen. The ventilators could not quite handle that load.

They stood huddled together, children to the inside, while those on the rim of the pack hugged their shoulders and clenched teeth between blue lips. Little was said. So far there was calm of a sort—enough personnel had had intensive mind training to be a steadying influence; but it was a thin membrane stretched near breaking.

As he came in, Gilchrist thought of a scene from Dante's hell. Somewhere in that dense mass, a child was sobbing. The lights were dim—he wondered why—and distorted faces were whittled out of thick shadow.

"G-g-get inside ... in front of me," he said to Catherine.

"I'll be all right," answered the girl. "It's a fact that women can stand cold better than men."

Alemán chuckled thinly. "But our Thomas is well padded against it," he said.

Gilchrist winced. He himself made jokes about his figure, but it was a cover-up. Then he wondered why he should care; they'd all be dead anyway, before long.

A colleague, Danton, turned empty eyes on them as they joined the rest. "Any word?" he asked.

"They're working on it," said Catherine shortly.

"God! Won't they hurry up? I've got a wife and kid. And we can't even sleep, it's so cold."

Yes, thought Gilchrist, that would be another angle. Weariness to eat away strength and hope ... radiation would work fast on people in a depressed state.

"They could at least give us a heater in here!" exclaimed Danton. His tone was raw. Shadows muffled his face and body.

"All the juice we can spare is going to the food and air plants. No use being warm if you starve or suffocate," said Catherine.

"I know, I know. But—Well, why aren't we getting more light? There ought to be enough current to heat the plants and still furnish a decent glow in here."

"Something else—" Gilchrist hesitated. "Something else is operating, then, and sucking a lot of power. I don't know what."

"They say the pile itself is as hot as ever. Why can't we run a pipe directly from it?"

"And get a mess of fast neutrons?" Catherine's voice died. After all ... they were being irradiated as they stood here and trembled.

"We've got batteries!" It was almost a snarl from Danton's throat. "Batteries enough to keep us going comfortably for days. Why not use them?"

"And suppose the trouble hasn't been fixed by the time they're drained?" challenged Gilchrist.

"Don't say that!"

"Take it easy," advised another man.

Danton bit his lip and faced away, mumbling to himself.

A baby began to cry. There seemed no way of quieting it.

"Turn that bloody brat off!" The tone came saw-toothed from somewhere in the pack.

"Shut up!" A woman's voice, close to hysteria.

Gilchrist realized that his teeth were rattling. He forced them to stop. The air was foul in his nostrils.

He thought of beaches under a flooding sun, of summer meadows and a long sweaty walk down dusty roads, he thought of birds and blue sky. But it was no good. None of it was real.

The reality was here, just beyond the walls, where Neptune hung ashen above glittering snow that was not snow, where a thin poisonous wind whimpered between barren snags, where the dark and the cold flowed triumphantly close. The reality would be a block of solid gas, a hundred human corpses locked in it like flies in amber, it would be death and the end of all things.

He spoke slowly, through numbed lips: "Why has man always supposed that God cared?"

"We don't know if He does or not," said Catherine. "But man cares, isn't that enough?"

"Not when the next nearest man is so far away," said Alemán, trying to smile. "I will believe in God; man is too small."

Danton turned around again. "Then why won't He help us now?" he cried. "Why won't He at least save the children?"

"I said God cared," answered Alemán quietly, "not that He will do our work for us."

"Stow the theology, you two," said Catherine. "We're going to pieces in here. Can't somebody start a song?"

Alemán nodded. "Who has a guitar?" When there was no response, he began singing a capella:

*"La cucaracha, la cucaracha,
Ya no quiere caminar—"*

Voices joined in, self-consciously. They found themselves too few, and the song died.

Catherine rubbed her fingers together. "Even my pockets are cold now," she said wryly.

Gilchrist surprised himself; he took her hands in his. "That may help," he said.

"Why, thank you, Sir Galahad," she laughed. "You—Oh. Hey, there!"

O'Mallory, off guard detail now that everyone was assembled here, came over. He looked even bulkier than before in half a dozen layers of clothing. Gilchrist, who had been prepared to stand impotently in the background while the engineer distributed blarney, was almost relieved to see the fear on him. *He* knew!

"Any word?" asked Catherine.

"Not yet," he muttered.

"Why 'ave we so leetle light?" inquired Alemán. "What is it that draws the current so much? Surely not the heaters."

"No. It's the pump. The air-intake pump down in the pile room." O'Mallory's voice grew higher. "It's working overtime, sucking in more hydrogen. Don't ask me why! I don't know! Nobody does!"

"Wait," said Catherine eagerly. "If the room's losing its warm gas, and having to replace it from the cold stuff outside, would that account for the trouble we're having?"

"No," said O'Mallory dully. "We can't figure out where the hydrogen's disappearing to, and anyway it shouldn't make that much difference. The energy output down there's about what it's supposed to be, you know."

Gilchrist stood trying to think. His brain felt gelid.

But damn it, damn it, damn it, there must be a rational answer. He couldn't believe they had blundered into an ugly unknown facet of the cosmos. Natural law was the same, here or in the farthest galaxy—it had to be.

Item, he thought wearily. The pile was operating as usual, except that somehow hydrogen was being lost abnormally fast and therefore the pump had to bring in more from Triton's air. But—

—Item. That couldn't be due to a leak in the heating pipes, because they were still at their ordinary pressure.

—Item. The gas in the pipes included some radioactive isotope. Nevertheless—

—Item. It could not be hydrogen-3, because the pile was working normally and its neutron leakage just wasn't enough to produce that much. Therefore, some other element was involved.

Carbon? There was a little methane vapor in Triton's atmosphere. But not enough. Anyway, carbon-13 was a stable isotope, and the pile-room conditions wouldn't produce carbon-14. Unless—

Wait a minute! Something flickered on the edge of awareness.

Danton had buttonholed O'Mallory. "We were talking about using the battery banks," he said.

The engineer shrugged. "And what happens after they're used up? No, we're keeping them as a last resort." His grin was hideous. "We could get six or seven comfortable days out of them."

"Then let's have them! If you thumb-fingered idiots haven't fixed the system by then, you deserve to die."

"And you'll die right along with us, laddybuck." O'Mallory bristled. "Don't think the black gang's loafing. We're taking the cold and the radiation as much as you are—"

"Radiation?"

Faces turned around. Gilchrist saw eyes gleam white. The word rose in a roar, and a woman screamed.

"Shut up!" bawled O'Mallory frantically. "Shut up!"

Danton shouted and swung at him. The engineer shook his head and hit back. As Danton lurched, a man rabbit-punched O'Mallory from behind.

Gilchrist yanked Catherine away. The mob spilled over, a sudden storm. He heard a table splinter.

Someone leaped at him. He had been an educated man, a most scientific and urbane man, but he had just been told that hard radiation was pouring through his body and he ran about and howled. Gilchrist had a glimpse of an unshaven face drawn into a long thin box with terror, then he hit. The

man came on, ignoring blows, his own fists windmilling. Gilchrist lowered his head and tried clumsily to take the fury on his arms. Catherine, he thought dizzily, Catherine was at least behind him.

The man yelled. He sat down hard and gripped his stomach, retching. Alemán laughed shortly. "A good kick is advisable in such unsporting circumstances, *mi amigo*."

"Come on," gasped Catherine. "We've got to get help."

They fled down a tunnel of blackness. The riot noise faded behind, and there was only the hollow slapping of their feet.

Lights burned ahead, Vesey's office. A pair of engineer guards tried to halt them. Gilchrist choked out an explanation.

Vesey emerged and swore luridly, out of hurt and bewilderment at his own people. "And we haven't a tear gas bomb or a needler in the place!" He brooded a moment, then whirled on Jahangir, who had come out behind him. "Get a tank of compressed ammonia gas from the chem section and give 'em a few squirts if they're still kicking up when you arrive. That ought to quiet them without doing any permanent damage."



The chief nodded and bounded off with his subordinates. In this gravity, one man could carry a good-sized tank.

Vesey beat a fist into his palm. There was agony on his face.

Catherine laid a hand on his arm. "You've no choice," she said gently. "Ammonia is rough stuff, but it would be worse if children started getting trampled."

Gilchrist, leaning against the wall, straightened. It was as if a bolt had snapped home within him. His shout hurt their eardrums.

"Ammonia!"

"Yes," said Vesey dully. "What about it?" Breath smoked from his mouth, and his skin was rough with gooseflesh.

"I—I—I—It's your ... y-y-your *answer!*"

They had set up a heater in his laboratory so he could work, but the test was quickly made. Gilchrist turned from his apparatus and nodded, grinning with victory. "That settles the matter. This sample from the pile room proves it. The air down there is about half ammonia."

Vesey looked red-eyed at him. There hadn't been much harm done in the riot, but there had been a bad few minutes. "How's it work?" he asked. "I'm no chemist."

Alemán opened his mouth, then bowed grandly. "You tell him, Thomas. It is your moment."

Gilchrist took out a cigaret. He would have liked to make a cavalier performance of it, with Catherine watching, but his chilled fingers were clumsy and he dropped the little cylinder. She laughed and picked it up for him.

"Simple," he said. With technicalities to discuss, he could speak well enough, even when his eyes kept straying to the girl. "What we have down there is a Haber process chamber. It's a method for manufacturing ammonia out of nitrogen and hydrogen—obsolete now, but still of interest to physical chemists like myself.

"I haven't tested this sample for nitrogen yet, but there's got to be some, because ammonia is NH_3 . Obviously, there's a vein of solid nitrogen down under the Hill. As the heat from the pile room penetrated downward, this slowly warmed up. Some of it turned gaseous, generating terrific pressure; and finally that pressure forced the gas up into the pile room.

"Now, when you have a nitrogen-hydrogen mixture at 500 degrees and 600 atmospheres, in the presence of a suitable catalyst, you get about a 45 percent yield of ammonia—"

"You looked that up," said Catherine accusingly.

He chuckled. "My dear girl," he said, "there are two ways to know a thing: you can know it, or you can know where to look it up. I prefer the latter." After a moment: "Naturally, this combination decreases the total volume of gas; so the pump has to pull in more hydrogen from outside to satisfy its barystat, and more nitrogen is welling from below all the time. We've been operating quite an efficient little ammonia factory down there, though it should reach equilibrium as to pressure and yield pretty soon.

"The Haber process catalyst, incidentally, is spongy iron with certain promoters—potassium and aluminum oxides are excellent ones. In other words, it so happened that the Hill is a natural Haber catalyst, which is why we've had this trouble."

"And I suppose the reaction is endothermic and absorbs heat?" asked Catherine.

"No ... as a matter of fact, it's exothermic, which is why the pile is actually a little hotter than usual, and that in spite of having to warm up all that outside air. But ammonia does have a considerably higher specific heat than hydrogen. So, while the gas in our pipes has the same caloric content, it has a lower temperature."

"Ummm—" Vesey rubbed his chin. "And the radiation?"

"Nitrogen plus neutrons gives carbon-14, a beta emitter."

"All right," said Catherine. "Now tell us how to repair the situation."

Her tone was light—after all, the answer was obvious—but it didn't escape Gilchrist that she *had* asked him to speak. Or was he thinking wishfully?

"We turn off the pile, empty the pipes, and go into the room in spacesuits," he said. "Probably the simplest thing would be to drill an outlet for the nitrogen vein and drop a thermite bomb down there ... that should flush it out in a hurry. Or maybe we can lay an impermeable floor. In any event, it shouldn't take more than a few days, which the batteries will see us through. Then we can go back to operation as usual."

Vesey nodded. "I'll put Jahangir on it right away." He stood up and extended his hand. "As for you, Dr. Gilchrist, you've saved all our lives and—"

"Shucks." His cheeks felt hot. "It was my own neck too."

Before his self-confidence could evaporate, he turned to Catherine. "Since we can't get back to work for a few days, how about going down to the bar for a drink? I believe it'll soon be functioning again. And, uh, there'll doubtless be a dance to celebrate later—"

"I didn't know you could dance," she said.

"I can't," he blurted.

They went out together. It is not merely inorganic reactions which require a catalyst.

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